

7.4 Detailed specific example: Mumbai, India



Located on the west coast of India, Mumbai is a rapidly growing megacity. Mumbai is growing at the rate of one person per minute, or half a million new people per year. Today, there are an estimated 25 million people living in the Mumbai Region.

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To accommodate the population, the city has grown outwards. Some of the outer suburbs are tens of km from central Mumbai. With limited funds, the big challenge facing the city authorities is how to manage this rapid growth without destroying the environment. In other words, how to manage Mumbai using sustainable strategies.



Source **A**: Population growth over time in Mumbai

► Image Description

Causes of growth

Mumbai's strategic coastal location helped it grow into a centre for international trade. Today, its expanding population is mainly due to inward migration. 60% of people moving to the city come from villages within the state of Maharashtra. Push factors from these villages include a lack of services such as safe drinking water, schools for children, medical facilities, and healthcare. Pull factors to Mumbai include the possibility of getting a better paid job in technology, banking or finance, fashion, or the film industry (Mumbai is home to Bollywood, the biggest film industry in the world).

TASK 1: Study Source **A**

- a** In which year did Mumbai become a megacity?
- b** In which decade did Mumbai's population increase most rapidly?
- c** Describe the population growth of Mumbai between 1950 and 2020.
- d** Explain three reasons for rural-urban migration to Mumbai.

Challenges, opportunities, and solutions

Rapid urban growth led to increasing numbers of rural to urban migrants building temporary housing. As a result, one of the world's biggest unplanned settlements, Dharavi (Figure 7.10), grew without any co-ordinated urban planning. An estimated one million people live there, in an area the size of about 500 football pitches. That is the equivalent of 2000 people per pitch. Residents now live in an informal settlement with inadequate services (water, sanitation, waste collection) and infrastructure (roads and transport) and worsening pollution leading to waterborne diseases such as cholera and typhoid. In such a densely populated city where land is scarce and land prices are rising, the government is very keen to redevelop this area into expensive real estate. However, 58 000 families live in Dharavi and there are over 12 000 businesses with over 100 000 people working in the huge informal economy, in industries like leather, ceramics, and textiles.



Figure 7.10: Dharavi, one of the world's largest unplanned settlements in Mumbai

To improve housing, the authorities have two choices:

- **bottom-up approaches:** where non-governmental organisations (NGOs) help residents improve the houses themselves and residents have a say in how things change
- **top-down approaches:** where the city authorities demolish the unplanned settlements and build new housing blocks, perhaps elsewhere in the city where there's more land available. These are called Slum Rehabilitation Schemes.

Governments have long debated how to revitalise Dharavi. The Dharavi Redevelopment project was originally proposed in 2004, but for a number of reasons was never carried out. In October 2022, new plans were developed to tackle the problems of informal housing in Dharavi. Companies were invited to propose plans and bid to redevelop the area. The Adani Group placed the winning bid and plans to work with the government to replace the informal housing with high-rise apartment blocks and better urban infrastructure. They will build free houses with water and power supply, gas pipes, and sewage disposal for some dwellers. An estimated 68 000 people (residents and business owners) would be relocated. To qualify for new housing, residents must prove that their informal housing was built before 1 January 2000.

This redevelopment is controversial. Many people live and work in connected spaces in Dharavi and buildings have been self-improved over time by residents. There is a strong sense of community between residents who have lived there for many years. Some residents would prefer to stay in their existing homes rather than move to an apartment block as they worry this may lead to a loss of community spirit and more crime. Many business owners also need to work at ground level and argue that high-rise apartments will destroy Dharavi's strong informal economy. Some residents and business owners complain that there has been little consultation and they don't know the plan details such as the size of rooms in the new apartments. This could be a big issue as there are thousands of extended families living within Dharavi who might only be offered one apartment.



Figure 7.11: Dharavi with high-rise tower blocks

TASK 2

Large-scale housing redevelopment schemes are controversial. Geographers need to identify and consider different viewpoints of those impacted by such schemes. Imagine that you are:

- a** a family currently living in Dharavi
- b** working for a property developer
- c** working for the city authorities

For each stakeholder position, think about your viewpoints about the proposed demolition and redevelopment of Dharavi. Discuss your ideas in small groups.

Transport

Mumbai's transport network hasn't been able to keep pace with its rapid urban growth. Public buses, private coaches, tuk-tuks, taxis, and an increasing number of private and Uber cars cause major congestion. During rush hour, traffic almost grinds to a halt. On many days, there's a constant pollution haze and the World Health Organisation ranked Mumbai as the fourth most polluted megacity in the world.

Mumbai's suburban train network covers 465 km and carries more than 7.5 million commuters every day. It is the busiest and most overcrowded rail network in the world. At rush hour, the trains are full of people trying to get to work on time. People often have to hang out of the doors (Figure 7.12). Two thousand

people die every year on Mumbai's railways from either falling from trains or being hit by trains as they cross the tracks.

Failure of Mumbai's monorail holds lessons for urban planners everywhere

It was hoped that India's first monorail would resolve Mumbai's transport woes – or some of them, at least. The MMRDA justified the monorail as best suitable for congested places since it requires less space, is capable of taking sharp turns and has the capacity to carry 20 000 passengers per hour per direction. The long-term purpose is to open up the land for development and spur economic growth. Some major residential projects by big builders have already begun construction in the area, which offers tremendous future possibilities.

Experts had warned even then that it was a foolhardy idea, which would have no practical impact on the city's transportation since monorail rakes have low carrying capacity, the planned route is unnecessary and expensive and the technology untested – it hasn't been used as a mode of mass transportation anywhere in the world. Additionally, cheaper modes, such as the bus rapid transit system – which could achieve the same ridership and speed at a lower cost – were not considered. It has suffered from a string of accidents, maintenance issues and stagnant passenger load. Its services were shut for ten months after a coach caught fire in November 2017. After the closure the monorail saw a decrease in ridership – from almost 15 000 per day to around 10 000. The monorail route was barely integrated with other modes of transport. Its stations are neither properly connected to suburban train stations nor are there frequent buses nearby to take passengers to their destinations. The closest suburban railway station where the monorail ends is 4 km away. Moreover, some stations are located away from residential areas. This increased the travel time and cost for passengers.

Source B: News article from *The Wire* about Mumbai's monorail project

TASK 3: Study Source B

Evaluate the strengths and weaknesses of the Mumbai monorail transport project as a strategy to improve the quality of life of people living in the city.

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Figure 7.12: Crowded commuter train in Mumbai

Part of the solution is a major urban public transportation project called the Mumbai Master Metro Plan (construction began in 2008). This is a huge engineering project linking business and residential areas in the south of Mumbai, the new banking and financial zone called BKC, the international airport, and major residential areas in the north of the city.

The metro system aims to carry 1.7 million passengers a day, reducing pressure on buses and taxis and cutting the number of private car journeys by around half a million per a day. Along its length, the project includes rainwater harvesting and solar panels to help reduce its ecological footprint. Once complete, it will connect with bus and pedestrian pathways to promote healthier living for users. The project, though, will take years to complete, is very expensive (costing an estimated \$4 billion) and will likely cause major congestion problems during construction.



Figure 7.13: The Mumbai monorail is the second largest monorail corridor in the world. It connects the eastern suburbs to central Mumbai and is expected to carry 200 000 commuters daily by 2030.

Green spaces

As Mumbai's population has grown over the last century from a few hundred thousand to over 20 million, the mangrove forest has been cleared to make way for housing and industry (Figure 7.14). Mumbai's growth means its green spaces are increasingly under threat. Some areas of mangroves are one of the few green spaces left in the city and they play a vital role in maintaining air quality. Mangroves moderate temperatures, reduce CO² levels, and act as filters to help keep rivers and coastal areas clean. They also provide spaces for people to relax. However, in 2017, the city's authorities announced a \$1 billion investment to update and expand the existing treatment plants, which meant cutting down 87 hectares of mangroves. There are campaigns to protect the mangroves of Mumbai and a proposal for a mangrove park in Dharavi.



Figure 7.14: A view of Powai lake, Mumbai

TASK 4

- a** Two of Mumbai's biggest challenges are air pollution and a lack of green space. In pairs, discuss which you think is the biggest challenge and why.
- b** In pairs, come up with a list of strategies that could help Mumbai meet the challenges of air pollution and a lack of green space.

IMPROVE THIS ANSWER

Describe and explain the impacts of urban sprawl on the rural-urban fringe.

Urban sprawl has many impacts on the rural-urban fringe. 1 There are positive and negative impacts. 2

The positive impacts are that large out-of-town shopping centres can be built on larger areas of cheaper land. 3 However, urban sprawl also increases pollution and congestion in rural-urban fringe areas. 4, 5

1 The answer should give a definition of urban sprawl and the rural-urban fringe here.

2 It is good that the answer has categorised impacts as good and bad, and links back to the question.

3 This point needs to be developed. Why is this a good thing? For example, linking to discussion of employment or easier access to services.

4 The answer needs to have some explanation here. Why does this happen? And it could further develop the negative impacts by linking these issues to health impacts.

5 The answer needs a conclusion.

Now, write an improved response to this answer using the guidance above.

SELF-EVALUATION CHECKLIST

After studying this topic, think about how confident you are with the different topics.

This will help you to find any gaps in your knowledge and learn more effectively.

I can:	Needs more work	Getting there	Ready to move on
describe and give reasons for differences in urban growth around the world			
understand the main causes of rapid urban growth in LICs			
identify the opportunities and challenges of urban living			
explain the challenges of rapid urban growth			
explain the impact of urban sprawl on the rural-urban fringe			
understand and evaluate the sustainable management strategies of urban growth.			

> Topic 8

The challenge of development

LEARNING INTENTIONS

By the end of this topic you will be able to:

- understand the different indicators of development
- evaluate strengths and weaknesses of development indicators
- describe and explain the global pattern of LICs, MICs, and HICs
- give reasons for differences in levels of economic development, standard of living, and the development gap between countries
- understand the importance of sustainable development
- evaluate strategies to address differences in levels of economic development.

GETTING STARTED

In pairs, discuss the following questions:

- 1 How do you think we can define poverty?
- 2 How do you think we can measure poverty?